R Practice 1 – Intro to R

This assignment is just for your own practice and does not need to be submitted. If you have any questions, please feel free to reach out to me, come to office hours, or schedule another a time to meet with me.

Open an R Script in RStudio and complete each of the following:

1. **Basic R Operations**
   a. Calculate the result of $8 - 3 \times 2$ in **R**. Provide the answer.

   b. Calculate the result of $3^4 + \sqrt{16} + e^3$**.** You can evaluate $\sqrt{16}$ using a function or exponent. Note that e is Euler's number and is approximately equal to 2.718. You can use the **exp()** function where the input is the exponent on e.

   c. Create a sequence of numbers from 1 to 15 with an increment of 2 using the **seq()** function.

   d. Create a vector containing the numbers from 5 to 15. Calculate the sum of all numbers in the vector.

   e. Given the vector **vec <- c(2, 4, 6, 8, 10)**, calculate the mean and median of the values in the vector.

2. **Creating Variables**
   a. Create a variable named **city** and assign the name of your favorite city to it. Print the value of the **city** variable.

   b. Create a variable named **birth_year** and assign your birth year to it. Print the message: "I was born in [birth year]." You can do this using the **paste()** function.

   c. Create a variable named **country** and assign the name of your home country to it. Print the value of the **country** variable.

   d. Create a variable named **birth_month** and assign the number corresponding to your birth month. Print the message: "I was born in month [birth_month]."

3. **Basics of Data Types**
   a. Create a character vector containing the names of three of your favorite books.

b. Declare a logical variable **is_employed** and assign a value to indicate your current employment status. Print the message: "Am I currently employed? [logical value]"

c. Create a numeric vector called **ages** containing the ages of 5 of your friends.

d. Create a vector called **fruits** containing the names of at least 3 different fruits, and a vector **prices** containing their corresponding prices (look these up online to get an estimate).

4. **Basics of Evaluating Functions**
   a. Given the function **ceiling(7.25)**, evaluate it and explain the result.

   b. Evaluate the function **abs(-7)** and explain what the function computes.

   c. Evaluate **round(3.14159, 2)** and explain the result.

   d. Remember that if you are unsure of what a function does, you can type a question mark in front of it to bring up documentation on the function. Type **?toupper** to determine what the function does and then use the function on the string "hello, world". Comment on what the functions does.

   e. Do some research and find two more built-in **R** functions that we have not seen yet in the notes or this homework and then evaluate them with something.

5. **R Libraries**
   a. Install and load the “MASS” package. Then load it using the **library()** function and use it to evaluate **fractions(2 + 1/4 + 8/7 + 10/3).**

   b. Install the **R** package “tidyverse”. This will take some time since it will install many other packages.

   c. Load the package “tidyverse”.